



2025 年 12 月 CSCA 数学中英双语试卷

选择题 (共 48 题)

1. 已知集合 $A = 1, 2$, 集合 $B = 1, 2, 3$, 则 (Given sets $A = 1, 2$ and $B = 1, 2, 3$, then)
- A. $B \subseteq A$
B. $A = B$
C. $A \in B$
D. $A \subseteq B$
2. 已知集合 $A = (-1, 3)$, 集合 $B = [2, 4)$, 则 $A \cap B$ 等于 (Given sets $A = (-1, 3)$ and $B = [2, 4)$, then $A \cap B$ equals)
- A. $[2, 3]$
B. $[2, 3)$
C. $(2, 3]$
D. $(2, 3)$
3. 二次不等式 $x^2 - 5x + 4 < 0$ 的解集是 (The solution set of the quadratic inequality $x^2 - 5x + 4 < 0$ is)
- A. $(-4, -1)$
B. $(-\infty, 1) \cup (4, +\infty)$
C. $(-\infty, -4) \cup (-1, +\infty)$
D. $(1, 4)$
4. 函数 $y = \frac{1}{\sqrt{x+1}}$ 的定义域是 (The domain of the function $y = \frac{1}{\sqrt{x+1}}$ is)
- A. $[-1, +\infty)$
B. $(-1, +\infty)$
C. $(0, +\infty)$
D. $(-\infty, -1)$
5. 若 a_n 是首项 $a_1 = 2$ 、公差 $d = 3$ 的等差数列, 则 a_{100} 等于 (If a_n is an arithmetic progression with first term $a_1 = 2$ and common difference $d = 3$, then a_{100} equals)
- A. 302
B. 305



- C. 299
- D. 298
6. 点 $(-1, 3 - \pi)$ 位于 (The point $(-1, 3 - \pi)$ lies in the)
- A. 第三象限 (third quadrant)
- B. 第一象限 (first quadrant)
- C. 第二象限 (second quadrant)
- D. 第四象限 (fourth quadrant)
7. 计算 $(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}) \cdot \tan \frac{\pi}{4} =$ (Compute $(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}) \cdot \tan \frac{\pi}{4} =$)
- A. $\frac{2}{\sqrt{2}}$
- B. 2
- C. $\sqrt{2}$
- D. 1
8. 下列函数为奇函数的是 (The following function is odd:)
- A. $y = x \sin x$
- B. $y = x^4$
- C. $y = \cos x$
- D. $y = x + x^3$
9. 函数 $y = x^3$ 的反函数是 (The inverse function of $y = x^3$ is)
- A. $y = \frac{x}{3}$
- B. $y = 3x$
- C. $y = x^3$
- D. $y = \sqrt[3]{x}$
10. 下列函数为单调递增函数的是 (The following function is monotonically increasing:)
- A. $y = (\frac{1}{3})^x$
- B. $y = \log_{0.5} x$



- C. $y = 2x + 1$
D. $y = \sin x$

11. 下列点位于第三象限的是 (The following point lies in the third quadrant:)

- A. $(-1, -2)$
B. $(-1, 2)$
C. $(1, 2)$
D. $(1, -2)$

12. 分式不等式 $\frac{x-1}{x-2} \leq 1$ 的解集是 (The solution set of the fractional inequality

$$\frac{x-1}{x-2} \leq 1 \text{ is})$$

- A. $(2, +\infty)$
B. $[1, 2]$
C. $(-\infty, 2)$
D. $(-\infty, 1]$

13. 已知 a, b, c 成等差数列, 且满足 $a + c = 20$, 则 b 等于 (It is known that a, b, c form an arithmetic progression and satisfy $a + c = 20$. Then b equals)

- A. 10
B. 5
C. 20
D. 0

14. 直线 $y = \sqrt{3}x + 10$ 的倾斜角等于 (The angle of inclination of the line

$$y = \sqrt{3}x + 10 \text{ equals})$$

- A. 60°
B. 30°
C. 45°
D. 120°

15. 下列不等式成立的是 (The following inequality is true:)

- A. $0.7^{-0.2} > 0.7^{-0.4}$
B. $2.1^{-2} > 1.2^{-2}$
C. $3^{2.25} < 3^3$
D. $2.1^{\frac{2}{3}} < 1.2^{\frac{2}{3}}$



16. 若点 $P(1, y)$ 在角 α 的终边上, 且 $\tan \alpha = 2$, 则 y 等于 (If the point $P(1, y)$ lies on the terminal side of an angle α and $\tan \alpha = 2$, then y equals)

- A. 1
- B. $\frac{1}{2}$
- C. $\sqrt{3}$
- D. 2

17. 已知 a_n 是等差数列, 且 $a_1 = 1$, $a_2 = 2$, 则 a_{2025} 等于 (It is known that a_n is an arithmetic progression and $a_1 = 1$, $a_2 = 2$. Then a_{2025} equals)

- A. 2
- B. 2025
- C. 2024
- D. 20

18. 若圆的圆心坐标为 $(-3, 2)$, 半径为 4, 则该圆的方程是 (If the center of a circle has coordinates $(-3, 2)$ and its radius equals 4, then the equation of the circle is)

- A. $(x + 3)^2 + (y - 2)^2 = 4$
- B. $(x - 3)^2 + (y + 2)^2 = 4$
- C. $(x - 3)^2 + (y + 2)^2 = 16$
- D. $(x + 3)^2 + (y - 2)^2 = 16$

19. 求点 $P(-1, -1)$ 与点 $Q(1, 1)$ 之间的距离 (Find the distance between the two points $P(-1, -1)$ and $Q(1, 1)$)

- A. 2



B. $2\sqrt{2}$

C. 4

D. $\sqrt{2}$

20. 求过点 $P(1,2)$ 和 $Q(2,4)$ 的直线方程 (Find the equation of the line passing through the two points $P(1,2)$ and $Q(2,4)$)

A. $y = x + 1$

B. $y = 2x + 2$

C. $y = x + 2$

D. $y = 2x$

21. 设圆的方程为 $x^2 + y^2 + 2x - 2y = 0$, 则该圆的半径等于 (Let the equation of a circle be $x^2 + y^2 + 2x - 2y = 0$. Then the radius of this circle equals)

A. 2

B. $\frac{1}{2}$

C. $\sqrt{2}$

D. 1

22. 若 α 为锐角, 且 $\sin\alpha = \frac{\sqrt{3}}{2}$, 则 $\tan\alpha$ 等于 (If α is an acute angle and $\sin\alpha = \frac{\sqrt{3}}{2}$, then $\tan\alpha$ equals)

A. $\frac{1}{2}$

B. $\frac{\sqrt{3}}{2}$

C. $\frac{1}{4}$

D. $\sqrt{3}$

23. 已知等比数列 a_n 的前三项为 1, 2, 4, 则该数列的通项公式是 (It is known that the geometric progression a_n has the first three terms 1, 2, and 4.



Then the general term of this progression is)

- A. $a_n = 2^n$
- B. $a_n = 8$
- C. $a_n = 2^{n-1}$
- D. $a_n = 2n$

24. 已知 $a > b$, 则 (It is known that $a > b$. Then)

- A. $\frac{1}{a} < \frac{1}{b}$
- B. $a^2 > b^2$
- C. $5^a > 5^b$
- D. $\sin a > \sin b$

25. 点 $P(-1,2)$ 到点 $Q(3,1)$ 的距离等于 (The distance from the point

$P(-1,2)$ to the point $Q(3,1)$ equals)

- A. $\sqrt{17}$
- B. $\sqrt{13}$
- C. $\sqrt{5}$
- D. 5

26. 关于函数 $y = \left(\frac{1}{2}\right)^x$, 下列说法正确的是 (The following statement about

the function $y = \left(\frac{1}{2}\right)^x$ is true:)

- A. 该函数关于原点对称 (This function is symmetric with respect to the origin)
- B. 该函数关于 x 轴对称 (This function is symmetric with respect to the x -axis)
- C. 该函数单调递增 (This function is monotonically increasing)
- D. 该函数单调递减 (This function is monotonically decreasing)

27. 若 $\sin \alpha = \frac{1}{4}$, 则 $\cos 2\alpha$ 等于 (If $\sin \alpha = \frac{1}{4}$, then $\cos 2\alpha$ equals)

- A. $-\frac{7}{8}$



B. $\frac{1}{8}$

C. $\frac{7}{8}$

D. $-\frac{1}{8}$

28. 直线 $l_1: y = 2x + 1$ 与直线 $l_2: x + y + 1 = 0$ 的交点坐标是 (The point of intersection of the line $l_1: y = 2x + 1$ and the line $l_2: x + y + 1 = 0$ is)

A. $(\frac{1}{3}, \frac{2}{3})$

B. $(\frac{2}{3}, \frac{1}{3})$

C. $(-\frac{1}{3}, -\frac{2}{3})$

D. $(-\frac{2}{3}, -\frac{1}{3})$

29. 双曲线 $x^2 - y^2 = 1$ 的焦点坐标是 (The coordinates of the foci of the hyperbola $x^2 - y^2 = 1$ are)

A. $(\pm\sqrt{2}, 0)$

B. $(0, \pm 2)$

C. $(0, \pm\sqrt{2})$

D. $(\pm 2, 0)$

30. $\log_2 32 + \log_{0.5} 32$ 的值等于 (The value of $\log_2 32 + \log_{0.5} 32$ equals)

A. -5

B. 10

C. 0

D. 5

31. 关于函数 $y = \cos x$, 下列说法正确的是 (The following statement about the function $y = \cos x$ is true:)

A. $\cos(x + \pi) = \cos x$

B. 该函数是奇函数 (This function is odd)



C. 该函数在区间 $[0, \pi]$ 上单调递增 (This function is monotonically increasing on the interval $[0, \pi]$)

D. 该函数的周期为 2π (This function is periodic with period 2π)

32. 关于抛物线 $y^2 = 4x$, 下列说法正确的是 (The following statement about the parabola $y^2 = 4x$ is true:)

A. 它经过点 $(-4, 4)$ (It passes through the point $(-4, 4)$)

B. 它的对称轴是 $x = 0$ (Its axis of symmetry is $x = 0$)

C. 它的焦点坐标是 $(\pm 1, 0)$ (The coordinates of its focus are $(\pm 1, 0)$)

D. 它的焦点坐标是 $(1, 0)$ (The coordinates of its focus are $(1, 0)$)

33. 下列直线与直线 $l: x + y + 3 = 0$ 垂直的是 (The following line is perpendicular to the line $l: x + y + 3 = 0$:)

A. $y = -2x + 1$

B. $y = x + 3$

C. $y = 1 - x$

D. $y = 2x + 1$

34. 已知 $\cos \alpha = -\frac{1}{2}$, $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, 则 $\sin \frac{\alpha}{2}$ 等于 (It is known that $\cos \alpha = -\frac{1}{2}$, $\alpha \in \left(\frac{\pi}{2}, \pi\right)$. Then the value of $\sin \frac{\alpha}{2}$ equals)

A. $-\frac{\sqrt{5}}{2}$

B. $\frac{\sqrt{3}}{2}$

C. $\frac{1}{2}$

D. $-\frac{1}{2}$

35. 已知 $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $\sin \frac{\pi}{6} = \frac{1}{2}$, 则 $\cos \frac{\pi}{12}$ 等于 (It is known that $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $\sin \frac{\pi}{6} = \frac{1}{2}$. Then $\cos \frac{\pi}{12}$ equals)

A. $\frac{\sqrt{6}}{4}$



B. $\frac{\sqrt{6}+\sqrt{2}}{4}$

C. $\frac{\sqrt{6}-\sqrt{2}}{4}$

D. $\frac{1+\sqrt{3}}{4}$

36. 下列函数与函数 $y = |x|$ 相等的是 (The following function coincides with the function $y = |x|$:)

A. $y = \sqrt[3]{x^3}$

B. $y = (\sqrt{x})^2$

C. $y = x$

D. $y = \sqrt{x^2}$

37. 若 α 为锐角, 且 $\cos\alpha = \frac{1}{2}$, 则 $\sin\frac{\alpha}{2}$ 等于 (If α is an acute angle and $\cos\alpha = \frac{1}{2}$, then $\sin\frac{\alpha}{2}$ equals)

A. $\pm\frac{\sqrt{3}}{2}$

B. $\pm\frac{1}{2}$

C. $\frac{\sqrt{3}}{2}$

D. $\frac{1}{2}$

38. 关于三角函数诱导公式, 下列公式不正确的是 (Regarding the trigonometric reduction formulas, the following formula is incorrect:)

A. $\cos(2\pi - \alpha) = \cos\alpha$

B. $\tan(\pi + \alpha) = -\tan\alpha$

C. $\sin(\pi - \alpha) = \sin\alpha$

D. $\sin\left(\frac{\pi}{2} - \alpha\right) = \cos\alpha$

39. 抛物线 $y^2 = -x$ 的准线方程是 (The equation of the directrix of the parabola $y^2 = -x$ is)

A. $x = \frac{1}{2}$



B. $x = -\frac{1}{2}$

C. $x = \frac{1}{4}$

D. $x = -\frac{1}{4}$

40. 已知 $\frac{\cos\alpha - \sin\alpha}{\cos\alpha + \sin\alpha} = \frac{1}{3}$, 则 $\tan\alpha$ 等于 (It is known that $\frac{\cos\alpha - \sin\alpha}{\cos\alpha + \sin\alpha} = \frac{1}{3}$. Then $\tan\alpha$ equals)

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{2}{3}$

D. $\frac{1}{3}$

41. 若数列 a_n 满足 $a_1 = 1$, 且 $a_n = \frac{1}{1 + \frac{1}{a_{n-1}}}$ ($n \geq 2$), 则 a_{100} 等于 (If the sequence a_n satisfies $a_1 = 1$ and $a_n = \frac{1}{1 + \frac{1}{a_{n-1}}}$ ($n \geq 2$), then a_{100} equals)

A. 50

B. $\frac{1}{100}$

C. 1

D. 100

42. 关于对数函数 $y = \log_a x$, 下列说法正确的是 (The following statement about the logarithmic function $y = \log_a x$ is true:)

A. 该函数的定义域为 $(-\infty, +\infty)$ (The domain of this function is $(-\infty, +\infty)$)

B. 当 $a > 1$ 时, 该函数在其定义域上单调递减 (For $a > 1$ it is monotonically decreasing on its entire domain)



C. 当 $a > 1$ 且 $x > 0$ 时, 恒有 $\log_a x > 1$ (For $a > 1$ and any $x > 0$ it holds that $\log_a x > 1$)

D. 当 $0 < a < 1$ 时, 该函数在其定义域上单调递减 (For $0 < a < 1$ it is monotonically decreasing on its entire domain)

43. 关于椭圆 $\frac{x^2}{4} + \frac{y^2}{20} = 1$, 下列说法正确的是 (The following statement about the ellipse $\frac{x^2}{4} + \frac{y^2}{20} = 1$ is true:)

A. 该椭圆的两个焦点坐标为 $(0, \pm 4)$ (The two foci of this ellipse are $(0, \pm 4)$)

B. 该椭圆的两个焦点坐标为 $(\pm 4, 0)$ (The two foci of this ellipse are $(\pm 4, 0)$)

C. 该椭圆的顶点坐标为 $(\pm 4, 0)$ 和 $(0, \pm 20)$ (The vertices of this ellipse are $(\pm 4, 0)$ and $(0, \pm 20)$)

D. 该椭圆的长轴长和短轴长分别为 20 和 4 (The lengths of the major and minor axes of this ellipse are 20 and 4, respectively)

44. 若一条直线与直线 $l_1: x + 2y + 4 = 0$ 垂直, 且经过直线 $l_2: x + y + 1 = 0$ 与 $l_3: 2x + y - 1 = 0$ 的交点, 则该直线的方程是 (If a line is perpendicular to the line $l_1: x + 2y + 4 = 0$ and passes through the intersection point of the lines $l_2: x + y + 1 = 0$ and $l_3: 2x + y - 1 = 0$, then the equation of this line is)

A. $x + 2y - 3 = 0$

B. $2x + y - 5 = 0$

C. $2x - y - 7 = 0$

D. $x - 2y - 1 = 0$

45. 设 O 为原点, A 、 B 为平面内两点, P 为线段 AB 的中点, 则向量



$\overrightarrow{OP}$ 等于 (Let O be the origin, A and B be two points in the plane, and

P be the midpoint of the segment AB . Then the vector $\overrightarrow{OP}$ equals)

A. $\frac{1}{2}(\overrightarrow{OA} + \overrightarrow{OB})$

B. $\overrightarrow{OA} + \overrightarrow{OB}$

C. $\overrightarrow{OB} - \overrightarrow{OA}$

D. $\frac{1}{2}(\overrightarrow{OB} - \overrightarrow{OA})$

46. 若复数 z 满足 $z^3 = 1$ 且 $z \neq 1$, 则 $1 + z + z^2 + \cdots + z^{999} =$ (If the

complex number z satisfies $z^3 = 1$ and $z \neq 1$, then $1 + z + z^2 + \cdots +$

$z^{999} =$)

A. $\frac{1+\sqrt{3}i}{2}$

B. $-\frac{1+\sqrt{3}i}{2}$

C. 1

D. $-i$

47. 已知等差数列 a_n 满足 $3(a_2 + a_6) + 2(a_6 + a_{10} + a_{14}) = 24$, 则前 13

项和 S_{13} 等于 (It is known that the arithmetic progression a_n satisfies

$3(a_2 + a_6) + 2(a_6 + a_{10} + a_{14}) = 24$. Then the sum S_{13} equals)

A. 104

B. 52

C. 39

D. 26

48. 十二生肖, 又称属相, 是中国用来纪年的十二种动物, 即: 鼠、牛、虎、

兔、龙、蛇、马、羊、猴、鸡、狗、猪。例如, 1984 年、1996 年、2008

年出生的人属鼠, 1985 年、1997 年、2009 年出生的人属牛, 以此类推。



随机选取三个人，至少有两个人属相相同的概率等于 (The twelve zodiac signs, also known as the animal symbols of the years, are twelve animals used in China to denote years, namely: Rat, Ox, Tiger, Rabbit, Dragon, Snake, Horse, Goat, Monkey, Rooster, Dog, and Pig. For example, people born in 1984, 1996, and 2008 were born in the Year of the Rat, and people born in 1985, 1997, and 2009 were born in the Year of the Ox, and so on. The probability that among three randomly chosen people at least two have the same zodiac sign equals)

A. $\frac{17}{72}$

B. $\frac{1}{12}$

C. $\frac{55}{72}$

D. $\frac{143}{144}$